SUSTAINABLE DESIGN

GILLETTE RAZOR CARTRIDGE

FRANCESCO RASTELLI



TRENDS AND MARKET

GLOBAL MARKET

GLOBAL STATISTICS

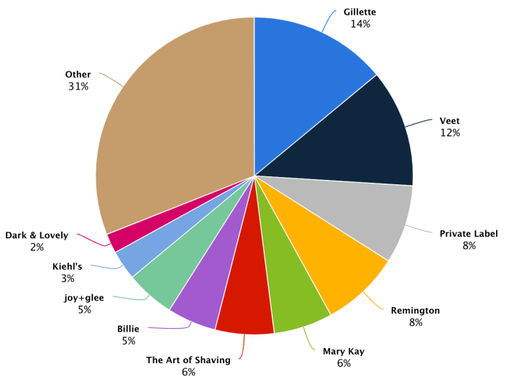
The worldwide revenue in the shaving market amounts to **US$35.37bn** in 2023. [1]

The market is expected to grow annually by **3.16%** (CAGR 2023-2028). [1]

In the UK, revenue during 2023 has been **US$0.50bn.** [1]

MARKET SHARES

Gillette has the biggest market share at **14%** [1]

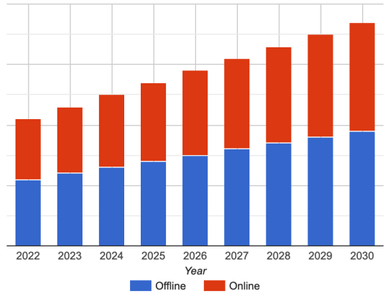


However, since 2010, the market share of the company in the US has fallen from **70** percent to **54** percent. [3]

The market is recovering though. Sales in 2022 increased by **3.2%** and a further growth of **3.6%** was expected in 2023 due to usage returning to pre-pandemic levels. [3]

THE SWAP TO ONLINE

The market share of Gillette has fallen in recent years, with other companies such as Harry's and Dollar Shave Club growing in size due to their ease of use of their subscription methods and cheaper average prices. The introduction of these alternative methods and with rising costs means more and more people are swapping to cheaper alternatives. [4]



WHAT THE EXPERTS SAY

"Value sales of shaving and hair removal products increased in 2022, with growth forecast to continue, driven by inflation. The category continues to battle relaxed attitudes toward hair removal, which have led to reduced usage frequencies. The cost of living crisis is further challenging the category as it is encouraging consumers to make savvy purchase decisions, but offering value beyond price point will help brands stand out. Additional opportunities lie within the consumer's heightened focus on skin health, making shaving and hair removal an essential step in self-care and skincare routines, with potential to expand product repertoires and drive higher value sales." Shiyan Zering Beauty and Personal Care Analyst [2]

GILLETTE

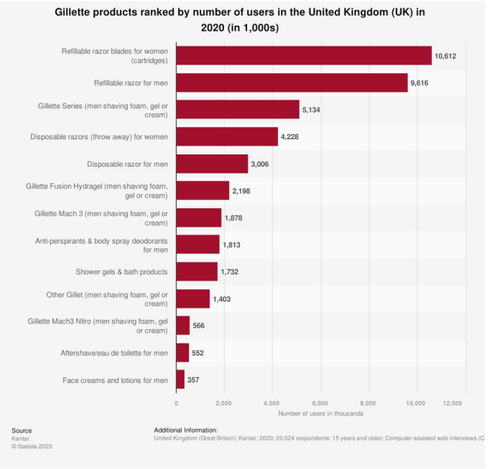
GILLETTE'S HISTORY [5]

1900 - Gillette's shaving history starts with their founder, King. Gillette, making the first prototype of the safety razor with a disposable blade.
1920 – Improvements to the design allow a more accurate blade angle and improve the grip.
1957 – The first adjustable razor with three settings was introduced for light, medium and heavy beards reaching up to 9 settings over time.
1965 – Techmatic is introduced which is the first razor with a continuous band that was spooled and retractable.
1972 – TRAC II is introduced which is the first twin blade system.
1977 – ATRA with its contour system arrives which can pivot and follow the contours of the face for an even better shave.
1985 - The lubricating strip is introduced with the ATRA PLUS.
1990 – The blades are then mounted on a spring meaning that they can adjust to the contours of the face even better.
1998 – MACH3 brings the first triple blade for an even smoother shave.
2006 - Fusion5 debuts with 5 blades granting some of the closest and most comfortable shaving.
2014 – FlexBall technology gives another pivot point allowing even better contour.
2015 – Skin guard allows for men with more sensitive skin to shave with less irritation.
2019 – Heated razors allow even more comfortability.



Gillette's Sales

In 2019 in the US, they generated over 150 million dollars in sales and sold over 13 million units. [3]



[6] Kantar. (2021). *Gillette products ranked by number of users in the United Kingdom (UK) in 2020 (in 1,000s)*. *Statista*. Statista Inc.. Accessed: October 20, 2023.

Consumer Behaviors

55% of UK consumers have removed body hair from their underarms in the last 12 months. [2]

50% of consumers agree that financial concerns would cause them to switch to a cheaper shaving/hair removal brand. [2]

PERSONAS AND USER RESEARCH

PERSONAS



Jorad

INFORMATION

AGE: 20
OCCUPATION: University Student
DEGREE: Mechanical Engineering

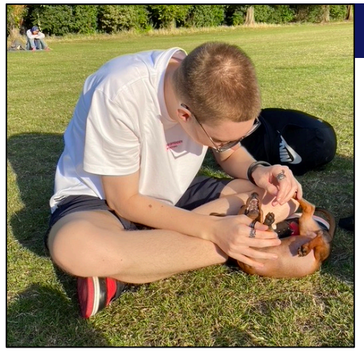
EXTRA CURRICULAR ACTIVITIES:
Partakes in Karate lessons and gyms.
LOCATION: Manchester

CORE NEEDS

- Shaves three times a week as his hair grows quickly and he likes the clean cut.
- Wants to shave quickly in the morning so needs the blades to cut well.
- Wants the blade to be as affordable as possible

PAIN POINTS

- His skin gets irritated easily from shaving
- He often finds that razors blunt quickly and pull on his hair instead of cutting them leading to an uncomfortable experience.
- Razors can be quite expensive



Jakub

INFORMATION

AGE: 21
OCCUPATION: University Student
DEGREE: Business and Management

EXTRA CURRICULAR ACTIVITIES:
Works in a cocktail bar.
LOCATION: Krakow

CORE NEEDS

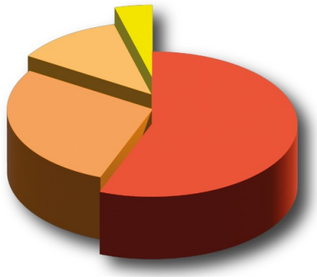
- Needs to have a clean shave for work even if it's part time so shaves regularly.
- He cares about the environment enough that he checks that the products he buys are eco-friendly.
- He prefers quality over cost but still looks for things that are affordable.

PAIN POINTS

- As he's often rushing between places, he tends to nick himself a lot when shaving.
- He often doesn't know whether the razors he uses are recyclable or not.
- He struggles to find non disposable razors where he lives.

USER RESEARCH

HOW OFTEN DO YOU SHAVE?

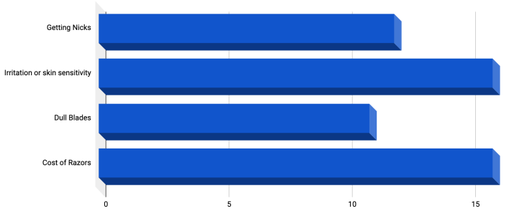


- **56%** - Once a week
- **28%** - 2-3 Times a week
- **12%** - Every day
- **4%** - Never

HOW IT WAS CONDUCTED

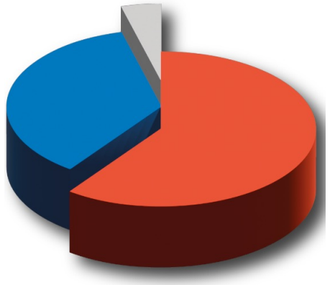
- A survey was run using Google Forms in order to get responses regarding certain key questions:
  - How often do you shave?
  - Are any of these issues with shaving for you?
  - Do you find razors to be too expensive?
  - Do you use disposable razors or refill razor cartridges?
- A total of 23 responses were gotten and the findings can be found on the left.

ARE ANY OF THESE ISSUES WITH SHAVING FOR YOU?



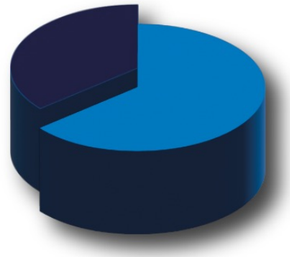
- **12** - Getting Nicks
- **16** - Irritation or skin sensitivity
- **11** – Dull Blades
- **16** – Cost of Razors

DO YOU USE DISPOSABLE RAZORS OR REFILL CARTRIDGES?



- **60%** - Refill Cartridges
- **36%** - Disposable Razors
- **4%** - I don't shave

DO YOU FIND RAZORS TO BE TOO EXPENSIVE?



- **64%** - Yes
- **28%** - No

USER JOURNEY MAP

JOURNEY MAP FOR JARED

The User Journey Map below is for Jared. It shows his experience from finding out about the Gillette razor head to purchasing and disposing of it. The height of the curve shows his user satisfaction and how it developed.

ACTIONS	VIEW	RESEARCH	PURCHASE	UNBOXING	USAGE	DISPOSAL
	Jared sees an ad for the refillable razor cartridges.	He then researches the product and decides to buy it.	He then goes online and orders it to his house.	The user receives and unboxes the cartridges.	Jared then starts using the product until it's time to change the head.	He then disposes and bins the product into his general waste bin.
SATISFACTION	'Curious / Interested'	'Driven / Planning / Exciting'	'Painful / Expensive / Exciting'	'Rushing / Inconvenient'	'Fast / Effective / Worry – Free'	'Confused / Careless / Simple'
PAIN POINTS	❑ The ad for the refillable razor cartridges makes them seem a lot eco-friendlier and as he's only used disposable razors this far, he decides to try them. ❑ He is curious about the benefits but decides to learn more.	❑ While researching the product, he looks through the Gillette website, online reviews on websites like Amazon, and word of mouth from friends to get an idea about weather this product is right for him.	❑ As Jared is used to purchasing stuff online, he decides to order it via Amazon and get it delivered straight to him for convenience.	❑ When the package arrives, he find that there is way too much wasted space and extra materials that did not need to be there. ❑ However, he is still eager to try it for his first shave.	❑ He then starts using the product and likes the clean shave it gives him. ❑ After a few weeks of usage, he notices the blade's gone dull and so decides to change it.	❑ When he has to dispose of it, he finds that he is not sure wether it can go into the recycling bin or not as it was not clearly labelled anywhere.

OPPORTUNITIES

Initial costs

Jared found the original "set-up" costs to be very expensive compared to the usual disposable razors that he'd buy.

This means that if we could reduce these costs, he would be more inclined to buy the Gillette branded one and so create a form of brand loyalty.

Packaging waste

The packaging was found to have a lot of wasted material in it which definitely impacts an eco friendly user's choice.

By reducing this, it would both benefit with reducing costs as the price of materials would decrease and it would also make our product more sustainable and eco-friendly.

Dulling of the blade

The heads only have to be binned when the blade dulls. This means that the longer you can make the blade last, the more sustainable it will become.

A blade that lasts longer also grows the brand loyalty to the product as it becomes more price friendly which Jared looks for as he is a university student and so often chooses the more affordable option.

Clearer recycling symbols

It was found that Jared was unsure of where the blade cartridges were supposed to be disposed. While he'd heard that they were supposed to be recyclable, he was still very unsure due to the mixed materials.

Clearer marking would make people more likely to recycle them thus increasing their sustainability.

PRODUCT TEARDOWN

DISSASSEMBLY PROCESS

Unboxing the packaging:
The packaging (1 – 3) was easy to disassemble using a craft knife and hands. From it, the razor and 2 razor heads were pulled out.

Unclicking the mount:
The mount (9) could be squeezed from both sides to push the clips in and pull it off the whole frame.

Disassembling the metal clips:
To get to the razors, the next step was to pull open the metal clips (6) that held the metal guard (6) and razors (5) to the plastic frame (7). To do this, pliers were used to straighten them and yank them out.

Removing the rubber guard:
From the razor head, the first thing that could be removed was the rubber guard (8) which was pried and pulled off.

Detaching the lubricating strip:
The lubricating strip (4) was glued on and so had to be disassembled using the craft knife to cut it off. It is also made up of two different plastics that were moulded together and so we were unable to disassemble them.

Removing the razors:
After this, the razors (5) could be pushed out from the back thereby finishing the disassembly.

COMPONENTS AND MATERIALS



PACKAGING (1 – 3)

- The main materials for the packaging is PP (1) and card (2 – 3) which are both easy to recycle.
- They were attached together using adhesives which although not the best, was still easy enough to separate.
- However, there was excess card put in (3) whose only purpose was to hide the empty space which greatly increases waste.

MAIN RAZOR HEAD (4 – 9)

- The razor head was made up of many different materials:
  - The frame (7) and mount (9) are made of PP.
  - The lubricating strip (4) is made of Polyurethane.
  - The rubber guard (8) is made of Siliocone elastomer.
  - Both the clips (6) and the razors (5) are made of high alloy stainless steel.
- Although not the easiest to disassemble, the lack of glue and ability to remove almost everything mechanically meant that it would be a lot easier to recycle.
- The use of a clear sequence of disassembly also made the whole experience easier.

REFLECTION OF THE PRODUCT TEARDOWN:

- **Design for disassembly:**
  - Overall, the product has been though off to be made to be disassembled with a clear order of how it should be taken apart and no permanent fixings (excluding the lubricating strip.
- **Design for recycling:**
  - All of the materials used within both the packaging and razor head are recyclable.
  - Gillette even has a recycling scheme with TerraCycle [7] which allows user to send in their razor heads via post for free and recycle them.
  - However, as shown by the user research, the lack of clear recycling symbols leads to confusion on whether the blades are actually recyclable or not.
- **Issues and Insights:**
  - The packaging seems quite wasteful compared to the size of the products inside it and could quite easily be reduced.
  - The number of different materials used could be reduced by changing the frame to the same type of stainless steel as the clips and razors.
  - The lack of clear recycling information can make the whole recycling process a lot more complicated than it actually has to be.

ECO AUDIT

ECO AUDIT

MATERIALS

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Energy (MJ)	%
Packaging (plastic)	Polypropylene (PP)	Virgin (0%)	0.013	1	0.013	0.89	63.6
Packaging (card)	Paper and cardboard	Virgin (0%)	0.01	1	0.01	0.29	20.8
Lubricating Strip	Polyurethane (tpPUR)	Virgin (0%)	9e-05	1	9e-05	0.0074	0.5
Razor Blades	Stainless steel	Virgin (0%)	0.0006	1	0.0006	0.044	3.1
Clips	Stainless steel	Virgin (0%)	0.00055	1	0.00055	0.04	2.9
Frame	Polypropylene (PP)	Virgin (0%)	0.0006	1	0.0006	0.041	2.9
Rubber Guard	Silicone elastomers (SI, Q)	Virgin (0%)	0.00022	1	0.00022	0.027	2.0
Mount	Polypropylene (PP)	Virgin (0%)	0.00087	1	0.00087	0.059	4.2
Total				**8**	**0.026**	**1.4**	**100**

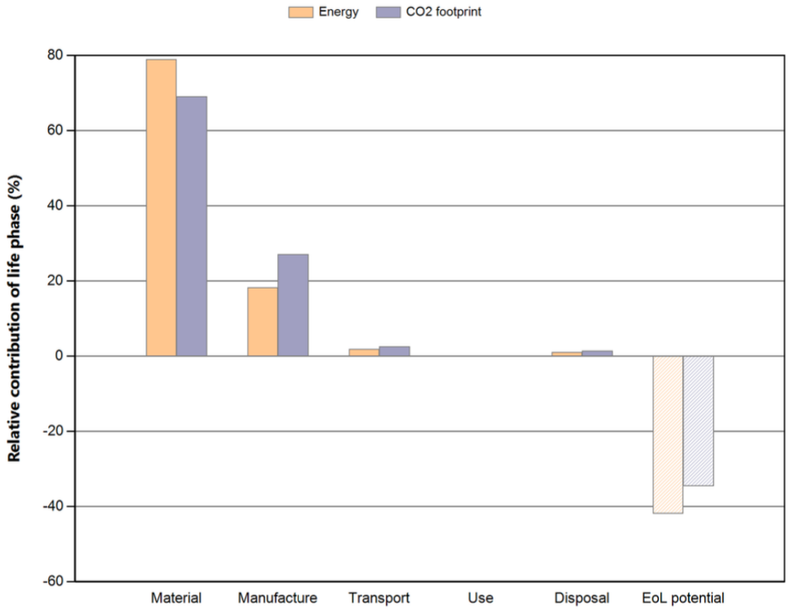
MANUFACTURING

Component	Process	Amount processed	Energy (MJ)	%
Packaging (plastic)	Polymer molding	0.013 kg	0.28	85.4
Lubricating Strip	Polymer molding	9e-05 kg	0.0017	0.5
Razor Blades	Extrusion, foil rolling	0.0006 kg	0.0059	1.8
Clips	Extrusion, foil rolling	0.00055 kg	0.0055	1.7
Frame	Polymer molding	0.0006 kg	0.013	3.9
Rubber Guard	Polymer molding	0.00022 kg	0.0033	1.0
Mount	Polymer molding	0.00087 kg	0.018	5.7
Total			**0.32**	**100**

DISPOSAL

Component	End of life option	Energy (MJ)	%
Packaging (plastic)	Recycle	0.0092	51.3
Packaging (card)	Recycle	0.0072	40.3
Lubricating Strip	Recycle	6.3e-05	0.4
Razor Blades	Recycle	0.00042	2.3
Clips	Recycle	0.00039	2.2
Frame	Recycle	0.00042	2.3
Rubber Guard	Landfill	4.4e-05	0.2
Mount	Landfill	0.00017	1.0
Total		**0.018**	**100**

COMPARISON OF ENERGY AND CO2 FOOTPRINT



ASSUMPTIONS

1. The product travelled from the factory in Berlin to London via truck travelling 1,104km.
2. The product is used for the 20 uses (that Gillette says it should last for before being changed) at an average of 2 shaves a week; thereby lasting 10 weeks or 0.192 years.
3. Users try to recycle as much as they can.
4. Additional products such as the handle, shaving cream and water usage have not been included in the calculations.

KEY MATERIALS AND TECHINQUES

Key materials

- **Stainless Steel:**
  - SS is 100% recyclable in theory; however, it is often not the case as a lot of it is disposed in general waste and so ends up in landfill. However, the growth of positive habits and the number of recycling plants will likely lead to almost all SS being recycled in the future.
- **PP:**
  - PP is widely recycled as it's a thermoplastic meaning that it can be melted down and re-softened in order to recycle it. However, the process is still quite expensive for its economic viability so not a lot of companies recycle it.

Key manufacturing techniques

- **Injection Moulding:**
  - Injection moulding is used for the production of the majority of the PP components. It is often used due to its ability to make many identical components for a relatively low cost and at high production output rates.
- **Metal Extrusion:**
  - The metal stainless steel components are almost all made through extrusion and sharpened by grinding. This is quite an energy intensive process, however, the use of mass production means that the costs are brought down together with quite a low material waste being generated which can just be re-melted down and recycled into new components.

Issues and opportunities

- **The packaging has one of the highest percentage of energy:**
  - The PP packaging makes up 63.6% of materials energy and 85.4% of manufacturing. Comparing that to the card components which are 20.8% and 0.5% respectively, it is much higher. Therefore, cutting down on PP in packaging would also decrease the energy cost.
- **Removing the rubber guard would make the razor 100% recyclable:**
  - The rubber guard is the only component of the razor that is not recyclable given it's a thermosetting plastic and so cannot be melted back down to be reshaped.
- **Reducing the number of parts:**
  - There is a large number of components which could be cut down thus decreasing energy use and CO_2 emissions given that multiple components could be made by themselves in a single process.

FLOW MAPPER

FLOW MAPPER

Resource

- **Product:**
  - Gillette Fusion 5 Disposable Razor Blades
- **Component:**
  - Razor Blades
- **Material:**
  - Stainless Steel

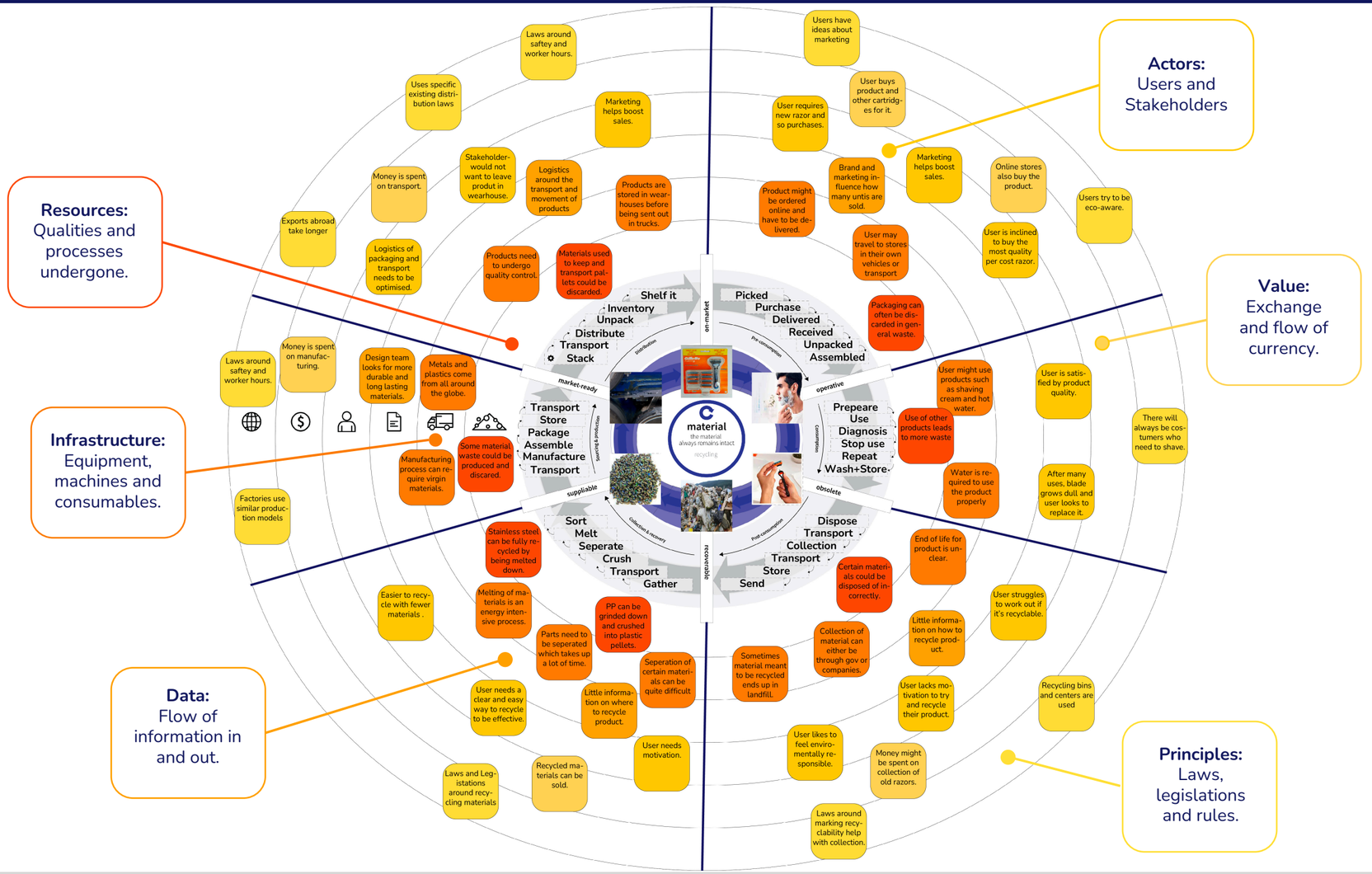
Consumer

- **What needs does the product fulfill?**
  - Removing hair. Giving a clean and comfortable cut.
- **What is the function of the product?**
  - Shaving of face and other body parts where hair grows.
- **Who is the key consumer?**
  - Men who shave often and like to have a clean cut face. Men with sensitive skin who don't want to irritate it.

Context

- **Which region is the product used?**
  - Worldwide especially in the US and Europe.
- **Where is the product used?**
  - Generally in the bathroom.
- **How often is it used?**
  - On average 2 to 3 times a week.

CONFIGURATOR BOARD



FLOW SYSTEM ANALYSIS

PROCESS	PROCESS SCOPE	PIVOTAL POINTS	RELEVANT ELEMENTS
Picked	The product is chosen on either a website or in a supermarket. The quality of packaging and marketing directly impacts the amount of products sold.	The better the marketing and the design, the more likely the costumer is to buy it as it will stand out to them more. This is beneficial to both the costumer and the stakeholders but can sometimes be frustrating for the costumer if they cannot find the item they were looking for.	Product might be ordered online and have to be delivered. User may travel to stores in their own vehicles or transport Brand and marketing influence how many untis are sold. Marketing helps boost sales. User is inclined to buy the most quality per cost razor.
Use	After the user puts the product together, they can begin using it. The quality of the experience will dictate if the user will keep on using this product,	The consumer needs to buy other products in order to use this properly like shaving gel, aftershave and access to clean water which is extra expenses for them.	Use of other products leads to more waste User might use products such as shaving cream and hot water. Water is required to use the product properly User is satisfied by product quality. There will always be costumers who need to shave.
Dispose	The product is thrown out when the blades dull and should end up in the recycling bin. Clear marking on the product incentives the user to recycle.	The razor is disposed of when the blade starts becoming dull and not cutting well anymore. This can take anywhere from a few week to a few months depending on how often someone shaves. When disposing of the razor, it can be quite difficult to know if it's recyclable or not given all the different materials within it.	Certain materials could be disposed of incorrectly. Little information on how to recycle product. User struggles to work out if it's recyclable. User lacks motivation to try and recycle their product. After many uses, blade grows dull and user looks to replace it.
Sort	The waste materials are sorted out to be recycled and turned into new products. Large amounts of different materials make this harder.	Given the large amount of different materials, it can be difficult to separate and sort them into materials that can be recycled and not. The lack of adhesives and permanent fixtures between materials does help but it is still difficult to recycle even if going through services like TerraCycle [7].	Seperation of certain materials can be quite difficult Parts need to be seperated which takes up a lot of time. Melting of materials is an energy intensive process. Easier to recycle with fewer materials . Recycled materials can be sold.
Package	The product is packaged and stored on pallets for transport. Plastic is used in the packaging process which often ends up not being recycled and often uses virgin materials.	Some material waste is produced as plastics are often used to hold the different boxes together for transport on pallets. These can often be discarded after use and not end up in a recycling plant and thus breaks the circular life system as virgin material needs to be added to the system each cycle.	Some material waste could be produced and discared. Metals and plastics come from all around the globe. Logistics of packaging and transport needs to be optimised. Laws around saftey and worker hours. Exports abroad take longer

RISK AND OPPORTUNITIES

RISKS

PACKAGING WASTE

- The main materials for the packaging is PP and card.
- Although these are easy to recycle, a lot of wasted material is being used and not all of the users end up throwing their packaging in a recycling bin.
- Although Gillette “introduced recyclable carboard boxes for its entire range of premium refillable razors” [8], some packaging is still found with plastic inside it.

LACK OF MOTIVATION TO RECYCLE

- Lots of users lack the motivation to recycle as a lot more effort is required to recycle the blades.
- This is especially true for a system like TerraCycle which requires users to mail in their old razor heads compared to just throwing them out of their home bin.
- A lot of people are aloo unaware of schemes like this existing as well which greatly reduced the number of people using them.

DULL BLADES

- The main reason for razors being thrown out is the blades dulling.
- Although they are supposed to last around 20 uses, this can vary from person to person and so can lead to them being thrown out early creating a lot more waste than is required.
- Being able to make your blades last longer would help alleviate this issue.

HARD TO RECYCLE

- There are 5 different materials found within the razor cartridge and head which can almost all be recycled if it wasn’t for the Silicone Elastomer found in the rubber guard.
- All these materials would end up at the recycling plant all together and can be difficult to separate and sort in a way that allows them to be recycled.

OPPORTUNITIES

REDUCE EMPTY SPACE

- One of the ways we could make this product more sustainable is by shrinking the size of the packaging so a lot less empty space exists.
- Using just card for the packaging would also help as reducing the number of different materials always increases the easiness of recycling.

RAZOR SHARPENING SCHEME

- Instead of trying to get more people to recycle, a way to make them more likely to send their products in is through a scheme where you would send your old razor heads in which could then be sorted and sharpened in order to make them like brand new without having to reuse old materials.
- This could be done for a much cheaper cost compared to buying new cartridges and so would be likely to motivate the user.

SHARPEN YOUR OWN BLADES

- Changing the design and introducing a way to sharpen your own blades would allow a user to use the same set of blades for a lot longer before having to throw them out.
- A way this could be done is by having the blades be removable and sharpened using something like a whetstone.
- Components like the lubricating strip could also be removed and replaced.

REDUCE NUMBER OF MATERIALS

- A lot of the frame materials, especially the PP, could be exchanged for materials such as stainless steel that are a lot easier to sort and recycle as they only need to be melted down.
- This would also increase the quality and strength of the product and make it possibly last longer. The issue is that components like the lubricating strip still need to exist, but this could be make removable and exchangeable.

REFERENCES

TRENDS AND MARKET

1. ‘*Shaving Market*’ Statista [cited Oct 2023]- Available from: https://www.statista.com/outlook/cmo/beauty-personal-care/personal-care/shaving/worldwide
2. ‘*UK Shaving and Hair Removal Market Report 2023*’ Mintel [cited Oct 2023] – Available from: https://store.mintel.com/report/uk-shaving-and-hair-removal-market-report
3. ‘*The Not so Smooth Future of the Global Razor Market*’ Andrew Menke – 27 Jan 2020 [cited Oct 2023] – Available from: https://globaledge.msu.edu/blog/post/56831/the-not-so-smooth-future-of-the-global-r
4. ‘*Global Shaving Market Insights*’ Skyquest – June 2023 [cited Oct 2023] – Available from: https://www.skyquestt.com/report/shaving-market
5. ‘*A PERSPECTIVE ON PRECISION*’ Gillette [cited Oct 2023] – Available from: https://gillette.com/en-us/about/our-story
6. ‘*Gillette products ranked by number of users in the United Kingdom (UK) in 2020 (in 1,000s)*’, Kantar – 2021 [cited 03 Nov 2023] – Available from: https://www.statista.com/statistics/312526/gillette-leading-products-in-the-uk/
7. ‘*The Gillette Free Recycling Envelope*’ Gillette [cited Oct 2023] – Available from: https://www.terracycle.com/en-GB/brigades/gillette-envelope-uk

PERSONAS AND USER RESEARCH

7. ‘*The Gillette Free Recycling Envelope*’ Gillette [cited Oct 2023] – Available from: https://www.terracycle.com/en-GB/brigades/gillette-envelope-uk

RISK AND OPPORTUNITIES

8. ‘*Gillette’s Recyclable Cardboard Packs*’ Gillette [cited Nov 2023] – Available from: https://www.gillette.co.uk/blog/shaving-science/sustainable-cardboard-packaging/